

A Highly Luminescent Metallo-Supramolecular Radical Cage

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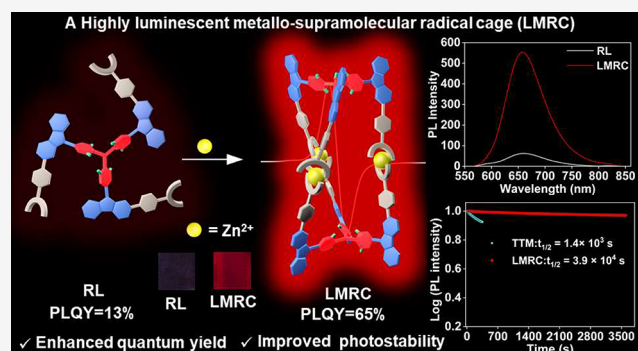


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ABSTRACT: Luminescent metal-radicals have recently received increasing attention due to their unique properties and promising applications in materials science. However, the luminescence of metal-radicals tends to be quenched after formation of metallo-complexes. It is challenging to construct metal-radicals with highly luminescent properties. Herein, we report a highly luminescent metallo-supramolecular radical cage (LMRC) constructed by the assembly of a tritopic terpyridinyl ligand RL with tris(2,4,6-trichlorophenyl)methyl (TTM) radical and Zn^{2+} . Electrospray ionization-mass spectrometry (ESI-MS), traveling-wave ion mobility-mass spectrometry (TWIM-MS), X-ray crystallography, electron paramagnetic resonance (EPR) spectroscopy, and superconducting quantum interference device (SQUID) confirm the formation of a prism-like supramolecular radical cage. LMRC exhibits a remarkable photoluminescence quantum yield (PLQY) of 65%, which is 5 times that of RL; meanwhile, LMRC also shows high photostability. Notably, significant magnetoluminescence can be observed for the high-concentration LMRC (15 wt % doped in PMMA film); however, the magnetoluminescence of 0.1 wt % doped LMRC film vanishes, revealing negligible spin-spin interactions between two radical centers in LMRC.



INTRODUCTION

Recently, luminescent radical materials have garnered a great deal of attention and interest from academia^{1–9} and have been widely applied in organic light emitting diodes (OLEDs),^{2,10,11} triplet sensitization,¹² magnetoluminescence,^{13–16} fluorescent sensors,^{17–20} circularly polarized luminescence,^{21,22} photodynamic therapy,²³ and so on. Most luminescent radical materials with high efficiency are organic radical molecules, including mono-, di-, tri-, and tetra-radicals.^{16,24,25} However, it is worth noting that the luminescence of metal-radicals tends to be quenched after formation of metallo-complexes.^{26–28} Thus, the development of luminescent metal-radical materials has been constrained in the past years.

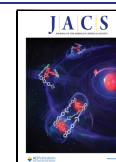
For breaking the limitation, researchers have employed the coordination-driven strategy with a highly directional, predictable, and reversible nature^{29–31} to construct a series of luminescent metal-radical complexes from zero-dimensional (0D) to three-dimensional (3D) in an “order-out-of-chaos” manner gradually by virtue of meticulously designed ligands and metal ions.^{14,28,32–41} In 2008, Wang et al. reported a 1D luminescent coordination polymer with Zn^{2+} , which is a pioneering breakthrough in metal-radicals with luminescent properties.³⁹ Veciana and co-workers constructed 3D porous metal-radical frameworks with Ln ions and investigated the luminescence and magnetic properties.²⁷ The Nishihara group explored 0D luminescent metal-radical complexes with PyBTM and Ag^+ ($\Phi_F = 0.08$)²⁸ and 2D magnetoluminescent

metal-radical polymers with Zn^{2+} .^{14,42} All the above studies have demonstrated that metal-radicals, which combine metal ions with radical motifs via coordination interactions elegantly, have also been proven to be an effective strategy for expanding the types of luminescent radical materials. However, to the best of our knowledge, the field of metal-radical cages with highly luminescent properties still remains unexplored at present.

Compared with other metallo-complexes, 3D metallo-cages with well-defined geometries and rigid topological architectures could restrict the conformations of organic motifs.^{43–51} This constraint has led to investigations into their luminescent properties in various fields, including magnetic resonance imaging,⁵² tunable and high emission,^{53,54} circularly polarized luminescence,^{55–58} and aggregation induced emission (AIE),^{59,60} by combining diverse functional groups. The study of metal-radical cages incorporating luminescent radical units will inject new vitality to the interdisciplinary frontier of metallo-supramolecular chemistry and luminescent radical materials.⁴¹

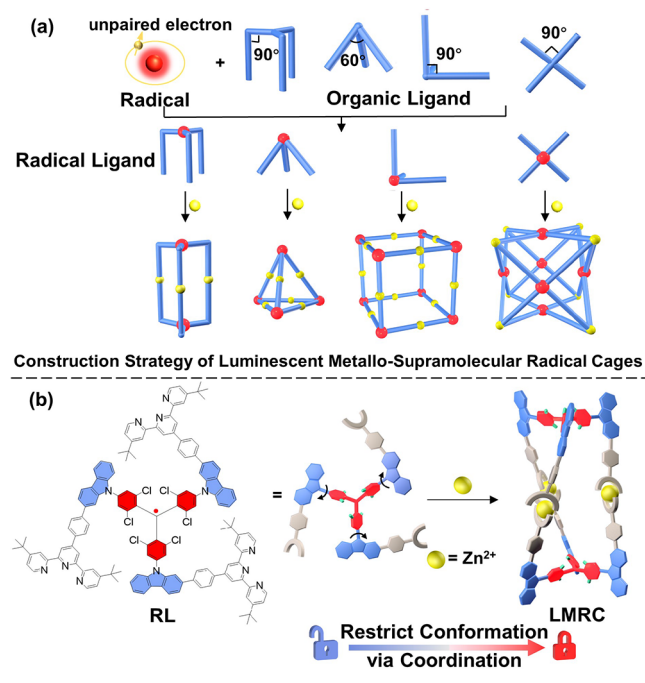
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Based on the versatile and predesigned nature of the ligand, it is feasible to construct novel luminescent radical cages by introducing functional motifs into organic ligands through metal-coordination interactions (Scheme 1a). In this

Scheme 1. Construction Strategy of Luminescent Metallo-Supramolecular Radical Cages (LMRCs) (a) and the Self-Assembly of the Trigonal Prism LMRC (b)



Article, we constructed a highly luminescent metallo-supramolecular radical cage (LMRC) by coordinating the tritopic terpyridinyl ligand RL with naked Zn^{2+} (Scheme 1b). Electrospray ionization-mass spectrometry (ESI-MS), traveling-wave ion mobility-mass spectrometry (TWIM-MS), and X-ray crystallography proved the successful formation of the twisted trigonal prism-like radical cage. Solid-state electron paramagnetic resonance (EPR) spectroscopy and superconducting quantum interference device (SQUID) analysis provided evidence of the existence of two radical centers in the obtained LMRC. The photoluminescent (PL) properties in solution and solid state were explored. LMRC can only be dissolved in high-polarity solvents, which caused PL quenching. In addition, a gradual enhancement in the PL intensity in 2-methyltetrahydrofuran was observed as the temperature decreased from 299 to 79 K. By doping in PMMA, the PLQY of LMRC reaches 65% and is 5 times that of RL (13%), which is higher than the highest PLQY of metal-radicals reported.³⁴ The magnetic field effects (MFEs) on luminescence were also explored. The magnetoluminescence of 15 wt % LMRC occurred, while the luminescence of 0.1 wt % LMRC was unaffected by the magnetic field, owing to the modulated spin-spin interaction. Additionally, LMRC showed a high photostability and thermal stability. All results implied that coordination-driven self-assembly played a vital role in constructing the highly luminescent metallo-supramolecular radical cage. The emergence of LMRC will not only signify a fundamental breakthrough in metallo-supramolecular radical cages but also broaden the scope of luminescent radical materials.

RESULTS AND DISCUSSION

Synthesis and Structural Characterization of LMRC.

The ligand RL is a tritopic terpyridinyl ligand with the “star luminescent radical” TTM radical⁵¹ and carbazole motifs. The electron-donating carbazole motif in RL provides the directing angles and plays a key role in adjusting the electronic density of LMRC. The precursor LH (Scheme S1) without radicals was synthesized through a Suzuki coupling reaction and characterized by ^1H , ^{13}C , and 2D COSY NMR (nuclear magnetic resonance) spectroscopy and MALDI-TOF (matrix-assisted laser desorption/ionization time-of-flight) mass spectrometry (Figures S2–S5). RL with TTM radicals was efficiently obtained from precursor LH by deprotection with $n\text{Bu}_4\text{N}^+\text{OH}^-$, followed by oxidation with p-chloranil, further purification with an Al_2O_3 column as a dark green powder, and characterization with MALDI-TOF MS (Scheme S1 and Figure S6). LMRC was obtained by self-assembling RL with $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in a stoichiometric ratio of 2:3 mixed in $\text{MeOH}/\text{CHCl}_3$ at 50 °C for 8 h, followed by the addition of excess NH_4PF_6 to give a dark green powder in quantitative yield (95%).

ESI-MS and TWIM-MS were utilized to determine the molecular composition of LMRC. In the ESI-MS spectrum (Figure 1a), only one prominent set of peaks with charge states

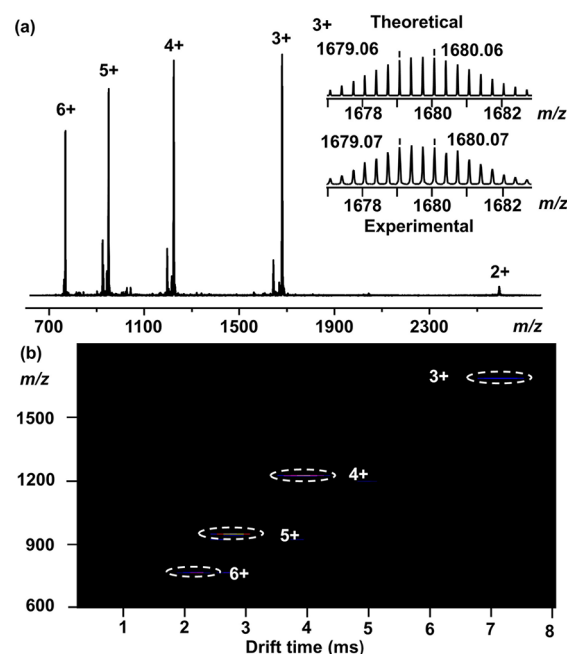


Figure 1. (a) ESI-MS spectrum and (b) TWIM-MS plot (m/z vs drift time) of LMRC. The charge states of the intact assemblies are marked.

from 2+ to 6+ was observed due to loss of different numbers of PF_6^- . The isotope pattern of each peak closely matched with the corresponding simulated isotope pattern of Zn_3RL_2 , consistent with the expected structure with a molecular weight of 5463.12 Da (Figure S7). The TWIM-MS spectrum (Figure 1b) showed a narrow drift time distribution of each charge state, indicating there are no other isomers or structural conformers.⁶²

To further elucidate the structural details, the diffraction-grade single crystal of LMRC as dark green squares was obtained by slow vapor diffusion of ethyl acetate into an

acetonitrile solution of LMRC and was definitively characterized by synchrotron radiation. The crystal structure of LMRC reveals it as a twisted trigonal prism composed of two RL and three Zn^{2+} (Figures 2a,c and S8), which belongs to the

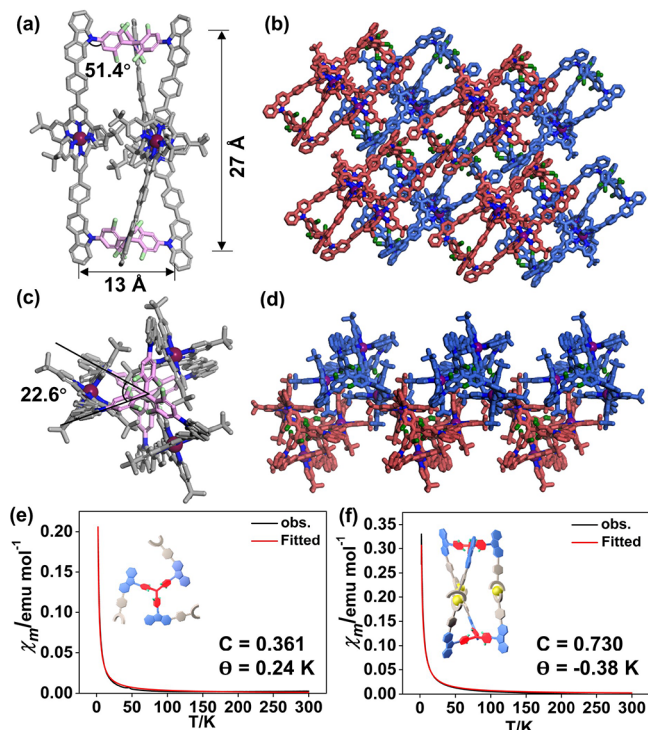


Figure 2. Single-crystal structure (a,c) and the packing mode (b,d) of LMRC; observed (black line) and fitted (red line) χ_m -T curves of RL (e) and LMRC (f).

triclinic system (Table S1). The space group is $P\bar{1}$, and a racemic pair of homochiral enantiomers was observed (Figures 2b,d, S9, and S10). The distance between two carbon radical centers is 27 Å, and the dihedral angle between the TTM radical and carbazole groups is approximately 51.4°. The three pyridyl rings of the TTM motif had a propeller-shaped conformation due to the steric hindrance, leading to the corresponding propeller-shaped conformation of carbazole rings and contributing to the chiral conformation of LMRC. The dihedral angle between the bottom surface and axis is 22.6° (Figure 2c), which confirms the formation of the twisted trigonal prism. The packing arrangement consists of one layer of cages with the *P* conformation (blue) and another layer with the *M* conformation (red) in Figure 2b,d. This packing arrangement is driven by CH- π interactions between two different enantiomers with distances ranging from 2.7 to 3.7 Å, as well as π - π interactions with a distance of 3.8 Å (Figure S9 and S10).

EPR was utilized to confirm the existence of spin $S = 1/2$ at room temperature. Both RL and LMRC exhibited a *g* value of 2.003 (Figure S11a,b), which is consistent with the characteristic *g* value of carbon radicals. To gain further insights into the magnetic properties of stable radicals in RL and LMRC, SQUID measurement was performed on powder samples in the temperature range of 2–300 K at an applied field of 500 Oe. From the χ_m -T curves of RL and LMRC (Figure 2e,f), the susceptibility reasonably follows the Curie–Weiss law. The best fit yielded a Weiss constant (θ) of 0.24 K and a Curie constant $C = 0.361 \text{ emu mol}^{-1}$ for RL, which nearly corresponds to the Curie constant of organic radicals $C = 0.375 \text{ emu mol}^{-1}$. For LMRC, a Weiss constant (θ) of -0.38 K and a Curie constant $C = 0.730 \text{ emu mol}^{-1}$ ($2 \times 0.365 \text{ emu mol}^{-1}$) were obtained, which correspond to the Curie constant of two radicals without interacting spin in LMRC, $C = 0.75 \text{ emu mol}^{-1}$ ($2 \times 0.375 \text{ emu mol}^{-1}$). These results affirmed that

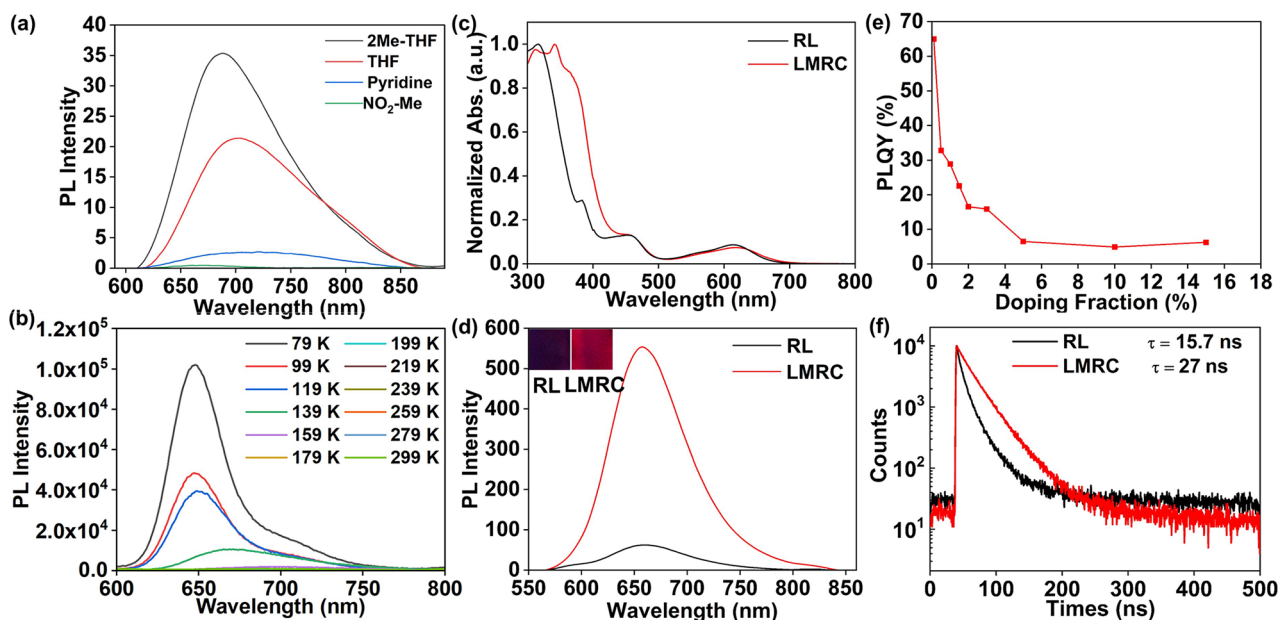


Figure 3. PL spectra of LMRC in different solvents (a) and 2-methyltetrahydrofuran at 79–299 K with an excitation wavelength of 375 nm (b) ($c = 10^{-6} \text{ mol/L}$); normalized UV–vis absorption spectra (c) and the PL spectra of 1.5 wt % RL and 1.5 wt % LMRC doped in PMMA (d) with an excitation wavelength of 375 nm (inset images at 365 nm UV light); PLQY of LMRC with different doping ratios in PMMA (e); time-resolved fluorescence decay profiles (f) of RL and LMRC.

RL and LMRC are paramagnetic and have negligible spin-spin interactions.

Luminescent Properties of RL and LMRC. After confirmation of the LMRC structure, the luminescent properties in both solution and solid state were further explored (Figures S12–S15). First, the PL spectra of LMRC in dilute solution were explored. In highly polar solvents (such as pyridine and nitromethane), the PL intensity of LMRC will be quenched (Figure 3a). For investigating the PL spectra of LMRC in dilute solution, we observed that LMRC has a luminescent band at 686 nm in 2-methyltetrahydrofuran. In addition, the PL spectra at different temperatures (299–79 K) were explored (Figure 3b) in 2-methyltetrahydrofuran; the PL intensity of LMRC is gradually increased, and the luminescent band displayed a blue shift as the temperature decreased. From the UV–vis (Figure S12), LMRC in solution and solid (with/without PMMA) states, the same characterized absorption band could be observed, proving that LMRC could exist stably in different states. The UV–vis absorption and PL spectra of RL and LMRC with 1.5 wt % doping in PMMA were measured. From the UV–vis spectrum (Figure 3c), RL displayed main absorption bands in the range of 300–400 nm as well as at 450 and 612 nm, originating mainly from charge transfer (CT) between terpyridine and carbazole and characteristic absorption of the TTM radical center. Compared with RL, the absorption band in the range of 300–400 nm of LMRC was bathochromically shifted by about 20 nm due to the introduction of coordination bonds. The absorption bands at 450 and 612 nm remained relatively unchanged. Both RL and LMRC exhibited emission bands at 660 nm in the PL spectrum (Figure 3d). Interestingly, the luminescence of LMRC was turned on and appeared bright red, in contrast to that of RL (inset images in Figure 3d), indicating that the radical center plays a major role in their emissions. The PL spectra of different temperatures and solid states indicated that the existence of the metal–coordination bond enhances the emission intensity of LMRC.

For further understanding the luminescent properties of RL and LMRC, PLQY and time-resolved decayed lifetime (τ : averaged value) were evaluated (Figures S16–S21, Table S2 and S3). The PLQY of LMRC (65%) is 5 times that of RL (13%) (Figure 3e and Table S2). The τ of LMRC is 27 ns, which is higher than that of RL (15.7 ns) (Figures 3f, S16, and S17). According to the results of PLQY and τ , the radiative decay rate (k_r) and nonradiative decay rate (k_{nr}) were investigated; k_r and k_{nr} were calculated to be $3.18 \times 10^7 \text{ s}^{-1}$ and $6.05 \times 10^7 \text{ s}^{-1}$ for RL and $8.14 \times 10^7 \text{ s}^{-1}$ and $2.88 \times 10^7 \text{ s}^{-1}$ for LMRC, respectively. Furthermore, the k_{nr}/k_r of LMRC was 3.5, which was much smaller than that of RL ($k_{nr}/k_r = 19$), indicating that under the action of coordination bonds, LMRC with a rigid backbone effectively inhibits the nonradiative transition process by restricting the intramolecular rotation or vibration.

To further explore the electronic structure of LMRC, we carried out the density functional theory (DFT) calculation at the UPBE0/6-31G (d) level in open-shell singlet and triplet states (Figures S23 and Figure 4). The spin density and frontier orbitals are illustrated in Figure 4. The spin density of LMRC is mainly localized at the radical center (Figure 4a), which indicates that two radicals exist independently in LMRC. Two α -singly occupied molecular orbitals (SOMOs) (1197 α , 1198 α) of LMRC are mainly delocalized on the TTM radical and extended to the carbazole group (Figure 4b).

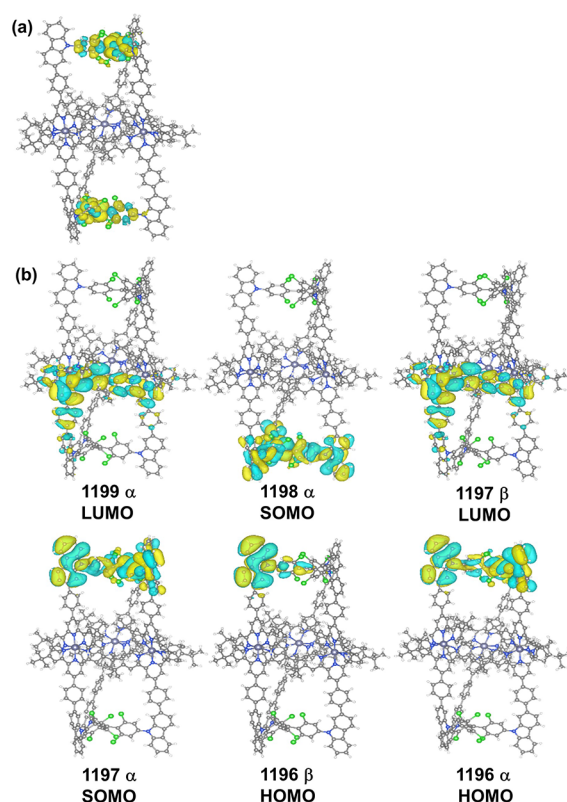


Figure 4. Calculated spin density (a) and singly occupied (α and β) orbitals (b) of LMRC calculated at the UPBE0/6-31G (d) level.

Conversely, the lowest unoccupied molecular orbitals (LUMOs; 1199 α and 1197 β) are distributed on the ligands of LMRC.

Considering the distance between two TTM radical centers (27 Å) of LMRC and the similar absorption and almost same emission bands of LMRC and RL, we conducted theoretical calculations on the fixed-conformation unit RL in LMRC and the free-conformation ligand RL using density functional theory (DFT) at the UPBE0/6-31G (d) level to further understand the frontier molecular orbitals and excited state properties of LMRC (Figure S24). The calculation results showed that the spin density of RL with a fixed conformation is mainly localized at the radical center (Figure S24a,b). The α - and β -highest occupied molecular orbitals (HOMO; 577 α and 577 β) are delocalized over the pyridine rings; however, the α - and β -singly occupied molecular orbitals (SOMO) (578 and 578 β) of RL are mainly delocalized on the TTM radical and extend to the carbazole group. All of the above results are consistent with free-conformation ligand RL (Figure S24d,e). The absorption spectra RL with fixed conformation matched well with the experimental result (Figure S24c), which was bathochromically shifted a little compared with that of RL with free conformation (Figure S24f). According to the time-dependent DFT calculation (TD-DFT) results, we can see that the transition from β -HOMO (577 β) to β -SOMO (578 β) is attributed to the transition of the doublet ground state (D_0) to the doublet excited state (D_1), which is related for emission. Combining the result of DFT calculations with the analysis of UV–vis and PL spectra, we think LMRC has doublet emission.

Luminescent Properties under the External Magnetic Field. Magnetic fields have the potential to affect fundamental molecular properties. The magnetic field effect (MFE) on

molecular luminescence has been extensively undertaken to gain insights into the spin dynamics and excited state characteristics of emissive molecules.^{13,63,64} For investigating the MFE on luminescence, the luminescent properties of LMRC under different external magnetic fields were explored. The emission spectra of 0.1 and 15 wt % LMRC doped in PMMA at 2 K under magnetic fields from 0 to 7 T are shown in Figure 5. For the emission spectra of 0.1 wt % LMRC

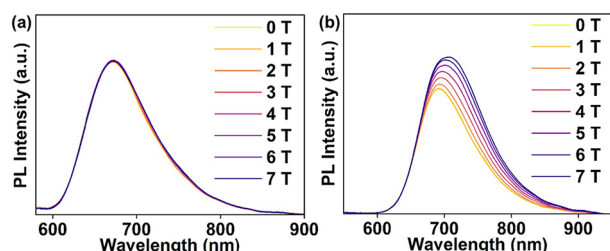


Figure 5. Emission spectra of 0.1 wt % (a) and 15 wt % (b) LMRC doped in PMMA at 2 K under a magnetic field with $\lambda_{\text{ex}} = 375$ nm.

(Figure 5a), the emission peak is observed at 670 nm, and the MFE on the luminescence (magnetoluminescence) could be negligible, which means that there is no spin-spin interaction of two radical centers in LMRC. However, magnetoluminescence became evident in the case of 15 wt % LMRC, and the emission-band peaks (690 nm) are distinctly emissive and exhibited a bathochromical shift of 15 nm under external applied magnetic fields. The emission intensity of LMRC increased by 25% with the magnetic field of 7 T. These results proved that the spin-spin interaction of two radicals of LMRC could be neglected in a low doping ratio. The distinction observed between low and high doping ratios of LMRC should be attributed to the modulation of radical–radical ground-state interactions, including factors such as the number of molecules and exchange interactions between molecules.¹⁴ The 3D rigid skeleton of LMRC could effectively control the intermolecular interactions by immobilizing the conformation and adjusting packing modes and distances between molecules.

Stability of LMRC. Stability is one of the most important parameters for evaluating the performance of luminescent radical materials. The thermal stability of LMRC was investigated with thermogravimetric analysis (TGA) under a nitrogen atmosphere (Figure 6a). The decomposition temperature (T_d corresponding to 5% weight loss) for LMRC was determined to be 284 °C. The photostability was also tested by monitoring the decay of emission intensities under continuous excitation at 375 nm. In comparison to the TTM radical

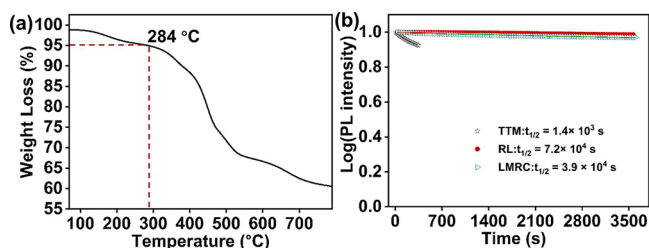


Figure 6. TGA data of LMRC (a) and the plots showing the PL intensity decays of 1.5 wt % TTM radical, 1.5 wt % RL, and 1.5 wt % LMRC doped in PMMA under 375 nm continuous photoexcitation (b).

(Figure 6b), LMRC and RL displayed excellent photostability; the estimated half-life time ($t_{1/2}$) of LMRC is 3.9×10^4 s, which is 26 times longer than that of the TTM radical. All the above data demonstrated that the LMRC has good stability.

CONCLUSION

In summary, the highly luminescent metallo-supramolecular radical cage (LMRC) was assembled with the TTM radical tritopic ligand RL and Zn^{2+} . LMRC was characterized by ESI-MS, TWIM-MS, and X-ray crystallography to conclusively give a Zn_3RL_2 twisted trigonal prism structure. EPR and SQUID results revealed that LMRC with two radicals exhibits paramagnetic behavior and weak spin-spin interactions. Both experimental and calculation results suggested that LMRC has doublet emission. Further investigation into the luminescent properties demonstrated that LMRC owns a high photoluminescence quantum yield (PLQY) of 65.0% and a longer average lifetime (τ) of 27 ns compared to RL. Meanwhile, LMRC displayed high photostability. The observed magnetoluminescence of 15 wt % LMRC, in contrast to the luminescence unaffected by the magnetic field of 0.1 wt % LMRC, could be attributed to the modulation of spin-spin interactions. This work presents a completely novel perspective and opportunities to develop highly luminescent radical materials.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.3c07477>.

Synthetic details and characterizations of ligands and complexes including NMR, MALDI-TOF MS, ESI-MS, TWIM-MS, single-crystal data, and data of luminescent properties (PDF)

Accession Codes

CCDC 2256742 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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